

Lexical Entailments as Partial structured explanations for Textual Entailments

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NLI task is popular, but...

We know the definition of the NLI task: $\langle P, H, l \rangle$.

One of the reasons of its popularity is being an easy task on reasoning:

- It is a three-way classification task.
- Simple/silly heuristics work due to annotation artifacts.
 - Hypothesis-only bias (Gururangan+ 2018, Poliak+ 2018, Tsuchiya 2018)
 - Word overlap bias (McCoy+ 2019, Naik+ 2018, Glockner+ 2018)
 - Inverse word overlap bias (Rajaei+ 2022)
 - Negation/antonymy bias (Lai&Hockenmaier 2014, Naik+ 2018)

Not every **dog** without a collar is barking **loudly**

E Some **animal** without a red collar is not barking

Motivation: labels are not explanations

Frontier models can solve many NLI instances, but the route to the answer is opaque.

NLI remains useful for probing specific reasoning phenomena

Free-text explanations (e-SNLI, Camburu+, 2018) are hard to evaluate automatically:

Prediction

Begins to climb a brick staircase means to go up the stairs

VS

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A dog begins to climb a brick staircase near plants.

E A dog is going up the stairs.

Brick staircase is a paraphrase of **stairs**, and **going up** means to **climb**.

Climbing implies **going up**.

A **DOG** IS CLIMBING UP THE **STAIRS**

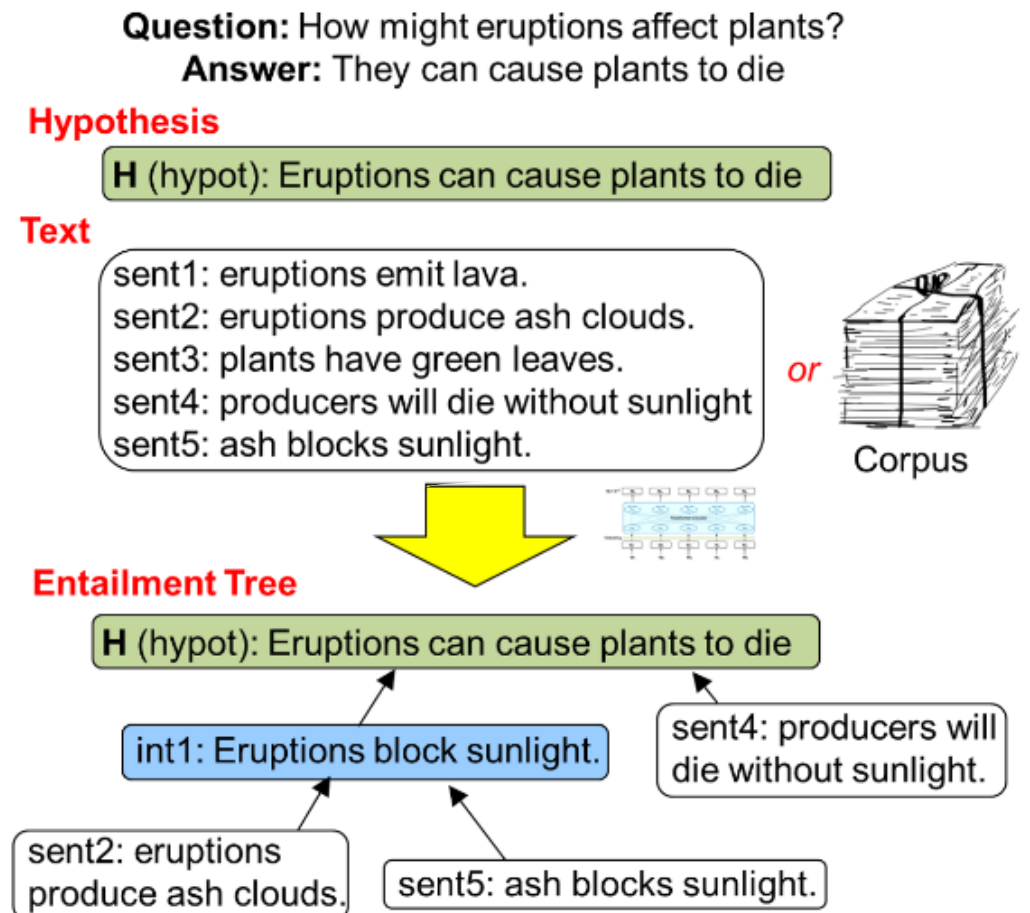
Symbolic provers give proofs, but they severely lack coverage

Structured explanations: limited & challenging

Structured explanations use a structure that limits the language variability in explanations and makes precise automated evaluation more tractable

Entailment trees: a seminal work on structured explanations for reasoning by Dalvi+ (2021).

It is not fine-grained.
Not applicable to existing NLI problems.



Research question and contributions

Can LLMs provide the right amount of lexical knowledge to explain and prove entailments?

1

Dataset

**NLI entailment pairs +
LEX explanations**

2

Intrinsic eval

**LLMs predict minimal
LEX sets**

3

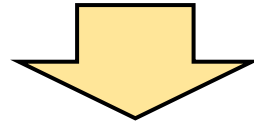
Extrinsic eval

**LLM KB improves
LangPro proofs**

LEX: structured lexical explanation

A zoo keeper holds a chinchilla.

E A person holds a small animal.



LEX set:

{ zoo keeper $\sqsubseteq$ person, chinchilla $\sqsubseteq$ small animal }

Directed relation (structured): first argument is more specific or equal to the second.

Not a full proof!

A compact proxy for the lexical knowledge needed.

Focus on entailment and minimality

Entailment label only

- Lexical relations often license entailments
- Lexical relations should be sensible out of the context too
- Neutral and contradiction are of a different sort: may involve negations and absence of relations

female swimmer $\sqsubseteq$ woman
~~swimmer $\sqsubseteq$ woman~~

Minimal LEX set

- No modifier deletion:
tall man $\sqsubseteq$ man is excluded
- Use relations over multi-word lexical units if single words are insufficient for explanation

A zoo keeper holds a chinchilla.	
E	A person holds a small animal.

chinchilla $\sqsubseteq$ small animal
~~chinchilla $\sqsubseteq$ animal~~

Curating NLI-LEX test set



Dataset statistic	Value
Total problems	362
SNLI / SICK	280 / 82
Unanimous agreement	223 (61.6%)
Alternative 4th LEX added	22 (6.1%)
LEX size dist. 0, 1, 2, 3+	29 / 51 / 14 / 6%

Examples: full agreement

Example A: no lexical bridge needed

P: A man in a bright yellow shirt juggles while riding a unicycle.

H: A man performs a juggling act on a unicycle.

All annotators

Example B: same non-empty LEX

P: A young boy wearing safety swimming gear and goggles is in a pool.

H: A kid uses eye protection while in the water.

All annotators

goggles $\sqsubseteq$ eye protection
pool $\sqsubseteq$ water
young boy $\sqsubseteq$ kid

The task includes both empty and non-empty explanations.

Examples: reference variation

P: A woman observes a rusted antique car.

H: A woman looks at an old car.

Annotator 1

antique $\sqsubseteq$ old
observe $\sqsubseteq$ look

Annotator 2

antique car $\sqsubseteq$ old car
observe $\sqsubseteq$ look

Annotator 3

antique $\sqsubseteq$ old
observe $\sqsubseteq$ look at

4th valid LEX

antique car $\sqsubseteq$ old car
observe $\sqsubseteq$ look at

Same inference, several acceptable lexical decompositions

Examples: references with disagreement

Example C: sand / beach

P: A person rides his bicycle in the sand beside the ocean.

H: A person is on a beach.

Annotator 1

sand $\sqsubseteq$ beach

Annotator 2 & annotator 3

Example D: dog / stick / field

P: A black and white dog is carrying a huge stick on the green grass.

H: A black and white dog with a large branch is running in the field.

Annotator 1

green grass $\sqsubseteq$ field
huge $\sqsubseteq$ large

Annotator 2

huge stick $\sqsubseteq$ large branch

Annotator 3

grass $\sqsubseteq$ field
huge $\sqsubseteq$ large
stick $\sqsubseteq$ branch

4th valid LEX

huge $\sqsubseteq$ large

Annotation variation is part of the phenomenon, not just noise.

Annotator performance

Annotator	n	Exact	micro-F1
A1 vs. rest	271	75.7	81.5
A2 vs. rest	271	84.5	86.0
A3 vs. rest	271	80.8	85.9
A4 vs. rest	273	82.8	86.2
Mean	–	80.9	84.9

Intrinsic evaluation: can LLMs predict LEX?

Data

- 362 curated entailment problems
- multi-reference gold LEX sets

Models

- 8 instruction-following LLMs
- hosted + open-weight comparison points
- 5 repeated runs

Evaluation

- micro-F1
- exact match for full LEX set
- no automatic pos-processing

Prompt choice:
annotation guidelines + few-shot
+ instructions specific to models' struggle

Intrinsic evaluation: results

Model	Micro-F1 $\pm$ SD	Exact match $\pm$ SD
Gemini 3.1 Flash-Lite	76.2 $\pm$ 0.6	65.4 $\pm$ 0.5
Claude Sonnet 4.5	75.9 $\pm$ 0.2	64.0 $\pm$ 0.2
GPT-5.4	74.1 $\pm$ 0.3	66.1 $\pm$ 0.9
Gemini 3.5 Flash	73.4 $\pm$ 0.5	60.9 $\pm$ 0.5
Claude Haiku 4.5	70.6 $\pm$ 0.0	57.2 $\pm$ 0.0
GPT-OSS-20B	61.5 $\pm$ 0.9	54.2 $\pm$ 1.1
GPT-5.4 Mini	59.3 $\pm$ 0.9	46.1 $\pm$ 1.1
Gemma 3 4B	31.2 $\pm$ 0.3	18.0 $\pm$ 0.3
Annot. avg.	84.9	80.9

Main pattern
model size alone does not explain performance
LLMs are underperforming

What makes LEX prediction hard?

The dominant tension

**Recover plausible relations
≠
select the minimal explanation**

- Recall generally exceeds precision
- Models add related but unnecessary relations
- Exact-match gap shows many partial predictions

Common error types

unnecessary

missing

reversed

granularity

normalization

Neuro-symbolic setup

For a textual entailment, LLM supplies a LEX set

A natural tableau-based theorem prover, LangPro (Abzianidze, 2017), uses the LEX set and proves the textual entailment



LLM does not predict the NLI label or construct the proof.

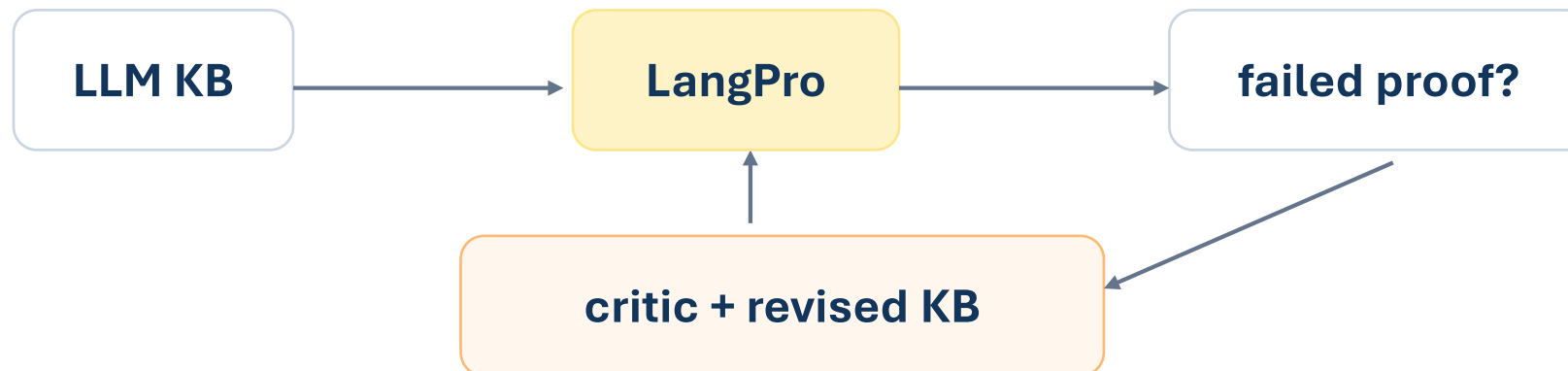
Extrinsic evaluation: do generated relations help proofs?

Evaluation set
1,000 SNLI entailment
problems

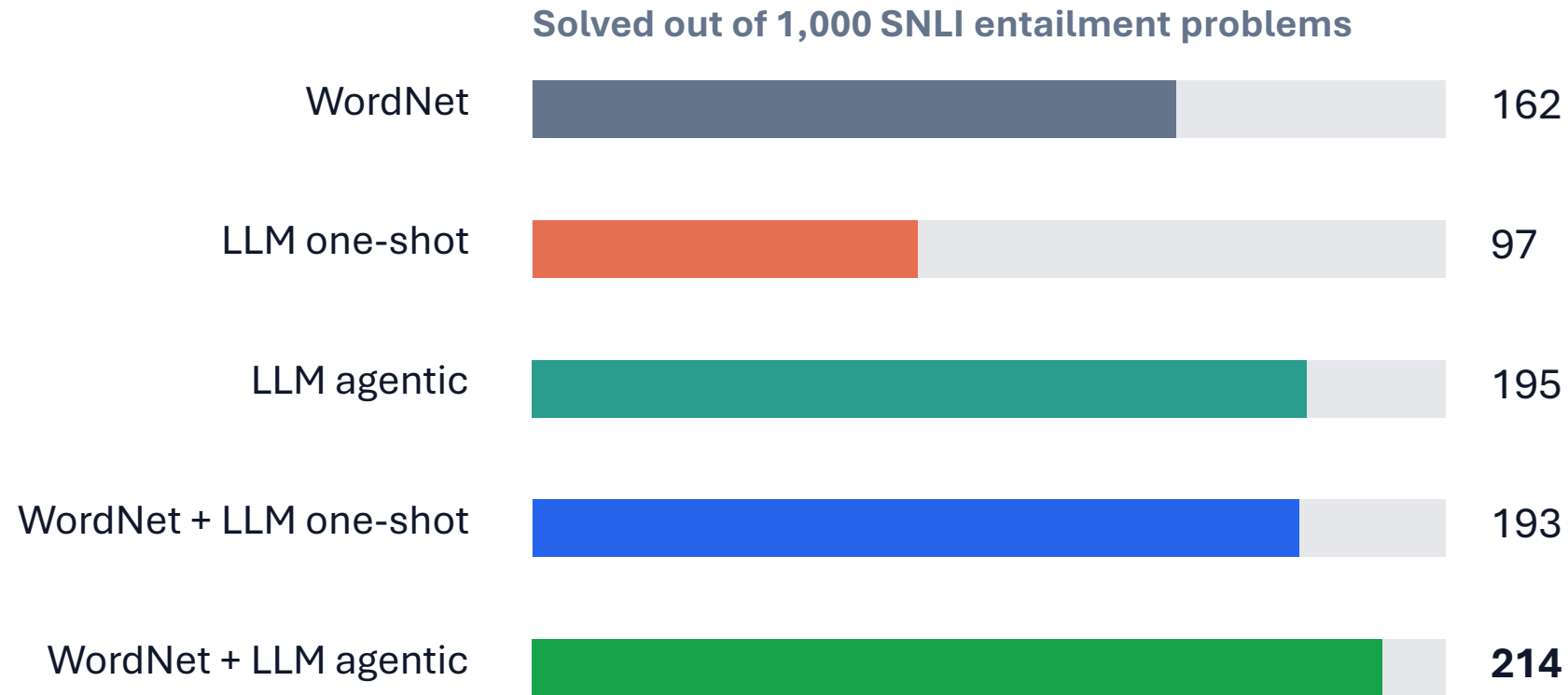
Generator
Gemini 3.1 Flash-Lite

Configurations

- WordNet only
- LLM only
- WordNet + LLM
- one-shot vs one refinement step



Proof coverage improves, but the gain is modest



+31 proofs
WordNet → WordNet+LLM one-shot

+21 proofs
after one refinement step

Limitations and caution

Scope limits

- entailment label only
- word and short-phrase relations only
- not complete explanations of NLI reasoning

Evaluation limits

- reference variants may not exhaust all valid LEX sets
- extrinsic results tied to LangPro representations

Takeaways

1

Structured LEX gives a compact, checkable explanation target.

2

LLMs recover much lexical knowledge, but minimality remains hard.

3

LLM-supplied KB slightly improves LangPro coverage.

4

We should put more effort in creating elaborated evaluation sets

Division of labor: LLMs supply flexible lexical candidates; symbolic systems perform explicit reasoning; intrinsic LEX evaluation checks validity.